Qihan Shan

shan6@engineering.upenn.edu | +86 158 5790 0687 | egg0523.github.io | github.com/Egg0523

EDUCATION

University of Pennsylvania Philadelphia, PA May 2025 – May 2027 (expected)
M.S. in Mechanical Engineering and Applied Mechanics — Mechatronics & Robotic Systems
Coursework: Robot Motion Control, Path Planning, Robot Kinematics, Machine / Deep / Reinforcement Learning

Zhejiang University–UIUC Institute (ZJUI) Sep 2021 – Jun 2025
B.S. in Mechanical Engineering, minor in Electrical Engineering | GPA 3.87 / 4.00

EXPERIENCE

AgiBot — Motion Control Algorithm Intern Shenzhen, China Jun 2026 – Present
X-Lab, Motion Intelligence Group | Layered navigation stack for the X2 humanoid

Semantic Planning — language-driven long-horizon navigation

- Built the navigation data pipeline end to end: collected 4,229 trajectories across 60 Matterport3D scenes, projected the 3-D paths onto egocentric views with depth-based occlusion filtering, and segmented them into 251,916 training samples.
- LoRA-finetuned a 7B vision-language model to emit goals as image-space coordinates, and trained a flow-matching trajectory network turning each goal into 32 dense waypoints; the two run asynchronously so planning never blocks control.
- VLN-CE RxR val-unseen: SR 59.3 / SPL 51.6. Projecting those goals into body-frame waypoints for the local planner closed the path from a language instruction to motion on the real X2.

Local Execution — on-robot navigation and dynamic obstacle avoidance

- Built a local planner around a learned dynamics model: proprioceptive history and an elevation map predict 5 s of future pose and failure risk, feeding a sampling-based planner that emits velocity commands with no hand-tuned cost maps.
- Deployed the full perception–planning–control pipeline on hardware, clearing doorways, pillar fields and corridors on flat ground and reaching goals up to 10 m away within 0.3 m.
- Built a 10 Hz perception–replanning loop for moving obstacles: 100% avoidance in simulation against a 1 m/s obstacle within 4 m, and 80% on hardware where sensing-to-actuation latency bites, the other 20% holding position rather than colliding — no collisions in either setting.

Perceptive Locomotion — traversing complex 3-D terrain (Isaac Lab)

- Trained an attention-based perceptive locomotion policy on an elevation map carrying per-cell uncertainty, distilling a privileged-map teacher into a student that runs on its own online mapping.
- Crossed stepping stones, stairs and slopes in both directions, and discrete raised platforms, arriving at each goal on the commanded heading: 90% teacher / 75% student success on unseen terrain.
- Retuned termination conditions and the terrain curriculum for an armed humanoid with a higher centre of mass and smaller support polygon; diagnosed and fixed foothold drift on consecutive stair ascents traced to odometry–elevation-map misalignment.

PROJECTS

UAV 3-D Path Planning and Real-Flight Deployment Feb – May 2026
UPenn | Advisor: Prof. M. Ani Hsieh

- Designed an SE(3) rigid-body pose controller with a low-level PD attitude loop, and combined A* global planning with piecewise quintic-polynomial fitting for smooth, collision-free, dynamically feasible trajectories.
- Integrated the full planning-to-control stack on hardware and flew autonomously through a real 3-D maze to a commanded landing point.

Quadruped Loco-Manipulation with a Mounted Arm Sep 2024 – Jan 2025
ZJU-UIUC Physical Intelligence Lab | Advisor: Prof. Hua Chen

- Led whole-body control modeling for dynamic mobile grasping, handling body posture, foot contacts, end-effector pose and stability constraints within a single policy.
- Trained PPO policies in massively parallel simulation with rewards spanning locomotion stability, manipulation, and coordination between the two.

SKILLS

Programming: Python, C, C++, MATLAB
Learning & Simulation: PyTorch, Isaac Lab / Isaac Sim, Habitat, MuJoCo; PPO, LoRA/PEFT, teacher–student distillation, diffusion policy and flow matching
Algorithms & Deployment: MPPI sampling-based planning, learned forward dynamics models, elevation mapping, VLN data pipelines, sim-to-real onboard deployment
Tools: ROS2, Linux, Git, SolidWorks, KiCad
Awards & Languages: Second Prize, National RoboMaster Robotics Competition 3v3, 2023 (ZJUI Meta RoboMaster Team) | Mandarin (native), English (TOEFL 103)